Suzuki triality geometries from generating cyclic triples

Human Compression project: L015 working application

This is an unreviewed complete-application draft for the fixed fourteen-page preprint of Leemans–Stokes–Tranchida [1]. We retain their triangle, chamber-system and coset constructions and their attribution to the classical Suzuki structure. The accompanying *Generating Suzuki triples by excluding proper subfields* supplies the uniform generating-seed theorem, including all its retained matrix and group-structure inputs. The entire four-page accepted *Coset trialities, chamber recognition and reduced relations* is a further common dependency; its Theorem 2 supplies the rank-three recognition argument. Neither common paper is a free import. No finite calculation or new formalization is a premise of this application.

1 Local supports and their stabilizers

Let $G = \mathrm{Sz}(q)$, $q = 2^n$, $n > 1$ odd, with its ovoid Ω. In addition to the explicit structure retained in the common paper we use the classical Zassenhaus property: no nonidentity element fixes three ovoid points. A nonidentity split-torus element fixes exactly its two endpoints; a nonidentity nonsplit-torus element fixes none, since a point stabilizer has odd part dividing $q-1$, coprime to the nonsplit torus orders. These are the source's classical action inputs, not new classification claims. For an odd dihedral subgroup $D = \langle x, y\rangle$ of order $2m$, $m > 1$, put $O(D) = \{\text{fixed point of } r : r \text{ is an involution of } D\}$.

Lemma 1 (Fixed-point and support recognition)**.** *Distinct involutions with the same fixed point commute. Distinct involutions with different fixed points have a product of nonidentity odd order. For fixed distinct points P, X and a fixed involution r at P, the products ru, as u ranges over the involutions at X, have pairwise distinct conjugacy classes. The support $O(D)$ has m points and determines D uniquely among dihedral subgroups of order $2m$.*

Proof. The involutions at a point are the nonidentity elements of its elementary abelian root center. Normalize P, X to $\infty, 0$, and write $r = z(b)$, $u = wz(t)w$ with $b, t \neq 0$. The common paper's direct trace formula gives $\mathrm{tr}(ru) = (bt)^\theta$. It is nonzero and varies injectively with t. The retained even-order trace property and invariance of trace prove both remaining fixed-point assertions. In particular two different reflections of D cannot share a fixed point, since their product is a nonidentity odd rotation. The cyclic rotation group $K = C_m$ acts regularly on the m reflections and hence on their fixed points.

Let $A = \mathrm{Stab}_G O(D)$. This is proper: m is at most the largest torus order, which is less than $q^2 + 1$. It contains D and cannot lie in a Borel, because D has involutions with different fixed points. Descend the maximal-subgroup list. If A equals a subfield Suzuki group $\mathrm{Sz}(r)$ with $r \geq 8$, then $O(D)$ lies in that group's subovoid and is a nonempty invariant subset. Transitivity makes it the whole subovoid, impossible since $m \leq r + \sqrt{2r} + 1 < r^2 + 1$. Thus descent eventually places A in an odd torus normalizer, with the terminal $\mathrm{Sz}(2) = C_5 : C_4$ included in that case. The rotation subgroup K is in the normalizer's unique odd cyclic subgroup, so that normalizer, and therefore A, normalizes K.

Let $T = C_G(K)$ be the ambient cyclic torus. We claim $A \cap T = K$. For $t \in A \cap T$ and $P \in O(D)$, regularity of K on the support lets us choose $k \in K$ such that tk fixes P. For a nonsplit torus this forces $tk = 1$. For a split torus, P is not an endpoint: a reflection inverting the torus interchanges its endpoints, since fixing both would put that involution in the odd two-point

stabilizer. The same conclusion follows. Hence A/K embeds in $N_G(K)/T$, which is cyclic of order two or four. Its unique order-two subgroup has preimage D. Any other dihedral group of order $2m$ with the same support lies in A; its odd rotations lie in $A \cap T = K$, and it has the same preimage. It therefore equals D. □

This proves the intended distinct-input version of source Lemma 2.2; the printed version must exclude $\zeta = \zeta'$ and require $P \neq Q$. The first omission gives identical products. For the second, take distinct ζ, ζ' at $P = Q$, both different from the fixed involution ω. Then $\omega\zeta, \omega\zeta'$ are nonidentity involutions and are conjugate: point double transitivity and the split-torus scaling show that all involutions form one class.

Fix an order-three automorphism τ and an involution a, put $b = \tau(a)$, $c = \tau^2(a)$, and suppose $|ab| = m > 1$ is odd. All pair products have order m. Write

$$H_0 = \langle b, c \rangle, \quad H_1 = \langle c, a \rangle, \quad H_2 = \langle a, b \rangle, \quad O_i = O(H_i).$$

The common paper's Sylow-conjugacy argument allows us to normalize τ to a field automorphism without changing any assertion here. The triple (O_0, O_1, O_2) is the source triangle: its pairwise intersections contain the fixed points of a, b, c, respectively. Cyclic symmetry implies that equality of any two supports, or any two H_i, implies equality of all.

Lemma 2 (Nondegeneracy and properness)**.** *The triangle is nondegenerate exactly when the H_i are distinct. In that case $H_i \cap H_j$ is the common marked involution subgroup. If $m \equiv 2 \pmod 3$, the triangle is nondegenerate. If $m = 5$, each pair of supports meets in just its marked fixed point, and their triple intersection is empty.*

Proof. The first claim follows from Lemma 1. If two different H_i share a nonidentity odd element, its centralizer is a torus containing both order-m rotation groups. A cyclic torus has only one subgroup of order m; the common marked reflection then makes the two H_i equal. If they share two distinct involutions, their product is such an odd element. Thus their intersection is precisely the common order-two subgroup.

If all H_i coincide, the argument in the common paper's descent lemma gives a unit u modulo m with $1 + u + u^2 \equiv 0 \pmod m$. A prime divisor of m congruent to two modulo three makes this impossible when $m \equiv 2 \pmod 3$. This proves nondegeneracy without assuming ambient generation.

For $m = 5$, suppose X is an additional point common to O_i, O_j, besides the point P of their common marked reflection r. Choose the respective reflections u, v fixing X. Both ru, rv have order five. There is a unique conjugacy class of order five, so the fixed-point trace injection forces $u = v$. This contradicts the just-proved intersection of the two dihedral groups. Thus the pair intersection is just P. The three marked fixed points are distinct, since each marked product has odd order greater than one; the triple intersection is therefore empty. □

The lemma recovers source Lemmas 3.4 and 3.6 without their use of the false unrestricted generation step. Nondegeneracy alone does *not* imply $\langle a, b, c \rangle = G$; that implication in source Corollary 3.5 is separately refuted by the common paper.

2 Prime products and the coset geometry

Lemma 3 (Prime rotation normalizers)**.** *If $K = C_p \leq G$ for an odd prime p, a dihedral subgroup D of order $2p$ contains K if and only if $D \leq N_G(K)$.*

Proof. Containment implies normalization because K is characteristic in D. Conversely $N_G(K)$ is its torus normalizer and has a unique subgroup of order p. The rotation subgroup of D must be that subgroup. □

Lemma 4 (The full four-element product identity)**.** *Suppose $m = p$ is prime and the source triangle is proper and nondegenerate. Then, with every ordered renaming of a, b, c allowed,*

$$H_1 \cap H_2 H_0 = \{1, a, c, ac\} = \langle a \rangle \langle c \rangle. \tag{1}$$

Proof. The displayed four elements plainly belong to the intersection. We prove the reverse containment. First take reflections $u \in H_2$, $v \in H_0$. If they coincide, their product is one. If they are distinct but have the same fixed point X, their product is another nontrivial root involution at X. Membership in H_1 would put X in all three supports, contrary to properness. With different fixed points their product is odd. If it belongs to H_1, it has order p and generates its rotation subgroup K_1. Both u, v invert uv, hence belong to $N_G(K_1)$. Also $a, c \in N_G(K_1)$. If $u \neq a$, the two distinct reflections u, a generate H_2 (here primality is essential), so Lemma 3 puts K_1 in H_2. The common reflection a then implies $H_1 = H_2$, a contradiction. Thus $u = a$; similarly $v = c$. The only reflection–reflection outputs in H_1 are consequently 1 and ac.

Next let $u \in H_2$ and $v \in H_0$ be nonidentity rotations. Rewrite $uv = (ub)(bv)$ as a product of two reflections. The result, if in H_1, is one or ac. It cannot be one, since this would give a nontrivial odd element in $H_2 \cap H_0 = \langle b \rangle$. Moreover equality to ac forces $ub = a$ and $bv = c$, so $u = ab, v = bc$. In particular a product of nontrivial rotations from two of the H_i cannot be a reflection in the third. This last assertion holds after every renaming of the indices.

Finally suppose $u \in H_2$ is a nontrivial rotation, $v \in H_0$ a reflection, and $z = uv \in H_1$. If z is a reflection, the identity $u = zv$ and the reflection–reflection argument, now with the indices renamed, force $z = a, v = b$, yielding $u = ab$ and output a. If z is a rotation, $u^{-1}z = v$ contradicts the rotation–rotation assertion. The reflected case gives output c. Products with an identity factor are covered by the pairwise intersections. These are all elements of the odd dihedral groups, proving (1). □

Theorem 5 (Selected generating triangles)**.** *Under Lemma 4, additionally suppose $\langle a, b, c \rangle = G$. Then $\Gamma = \Gamma(G; H_0, H_1, H_2)$ is a thin, residually connected, flag-transitive rank-three geometry with full typed automorphism group G. Every rank-two residue is a p-gon.*

Proof. Lemma 2 gives the three pairwise order-two intersections and trivial triple intersection. Equation (1) gives the exact product identity of Theorem 2 in the whole retained common paper on coset trialities. That theorem proves the geometry, all-flag transitivity, thinness, residual connectivity and full typed group, with no finite enumeration. The rank-two residue at H_i is the coset graph of D_{2p} over its two marked reflection groups, hence a cycle of length $2p$, a p-gon. □

Without $\langle a, b, c \rangle = G$, the same argument applies to the generated subgroup H, and the ambient coset system is a disjoint union indexed by the cosets of H. This is the exact missing generation boundary in source Theorem 3.14 and the final direct-coset paragraph. The subgroup product calculation does not by itself connect those components.

3 The literal triangles and chamber-system identification

Lemma 6 (The three-point obstruction)**.** *Let P, Q, R be three distinct ovoid points cycled by τ. Their setwise stabilizer in G is trivial.*

Proof. The pointwise stabilizer is trivial by the Zassenhaus property, so the setwise stabilizer embeds in S_3. It contains no three-cycle because $3 \nmid |G|$. If g induces a transposition, its image under conjugation by τ induces a different transposition. Their product is a forbidden three-cycle. The stabilizer is therefore trivial. □

For a nondegenerate triangle, an element of G stabilizing its unordered set of three supports permutes the three H_i, by Lemma 1, and therefore permutes their pairwise marked involutions, by Lemma 2. It stabilizes their three fixed points and is the identity by Lemma 6. Thus both the

ordered and unordered triangle orbits have size $|G|$, proving the full source Corollary 3.10. The distinct-cycle interpretation is essential in source Lemma 3.8.

Use left action notation temporarily: g carries the base triangle T to gT, and carries a subgroup H_i to gH_ig^{-1}. The fixed-point sets transform in exactly the same way, proving source Lemma 3.9 with its conjugated pair $(gag^{-1}, g\tau g^{-1})$. The unique reflection attached to type i at gT is gr_ig^{-1}, where $(r_0, r_1, r_2) = (a, b, c)$. The source adjacency rule therefore joins gT to $(gr_ig^{-1})(gT) = gr_iT$. It is exactly right multiplication by r_i on the chamber labels G. This proves symmetry, the two-element panels, equivariance and source Theorem 3.12 without a hidden connectedness premise. Its components are $g\langle a, b, c\rangle$.

The type-i objects of the associated chamber incidence system are the components under the other two colors, namely gH_i. Two such objects are incident exactly when these subsets meet. Inversion sends these left cosets to the right cosets H_ig^{-1} of Theorem 5; it preserves nonempty intersections. This is an explicit identification with the source geometry, rather than only an equality of diagrams or groups.

4 Full correlations and hypermaps

Proposition 7 (Normalized correlations)**.** *Let a thin geometry have full typed group G acting regularly on chambers. Fix a chamber C, and write r_i for the unique group element sending C to its other type-i panel chamber. Every correlation with type permutation π induces an automorphism of G sending r_i to $r_{\pi(i)}$. If its chamber graph is connected, the normalized chamber stabilizer embeds in the permutation group on the types.*

Proof. Multiply the correlation by the unique typed automorphism that returns its image of C to C. Call the result β. The typed group is the kernel of the type action, hence is normalized by β. Conjugation by β is thus an automorphism of G. The element $\beta r_i\beta^{-1}$ sends C to its unique other type-$\pi(i)$ panel chamber, so is $r_{\pi(i)}$. This proves source Proposition 3.15 without assuming the chamber graph connected or asserting that these r_i already generate G. In the connected case, a typed automorphism fixing C fixes its unique neighbors and then every chamber along all paths; this proves the claimed embedding. The source's preceding assertion that all r_i generate G needs this connectedness hypothesis; the proposition itself does not. □

Theorem 8 (The complete main geometry conclusion)**.** *For every odd $n > 1$ divisible by three, $G = \mathrm{Sz}(2^n)$ is the full typed automorphism group of a thin, residually connected, flag-transitive rank-three geometry with each of the three rank-two residues a pentagon. Its full correlation group is $G \rtimes C_3$, with trialities and no dualities.*

Proof. Choose the full generating seed of order five from the uniform common theorem. Lemma 2 makes its triangle proper and nondegenerate. Theorem 5 supplies every claimed geometric property and the full typed group. The actual order-three field automorphism sends H_i to H_{i+1}; on right cosets it acts as $H_ig \mapsto H_{i+1}\tau(g)$. The first proposition of the common coset paper proves this is a correlation of order exactly three.

Normalize any correlation to fix the standard chamber. Its type action is faithful, by Proposition 7. If a transposition occurs, the corresponding normalized correlation β has square the identity. Conjugation on G is then an automorphism of order dividing two and interchanges two of the distinct marked involutions. Since $\mathrm{Out}(G) = C_n$ has odd order, this automorphism is inner. Its conjugating element permutes the three marked fixed points, so is the identity by Lemma 6, a contradiction. The type image contains C_3 and no transposition, hence equals C_3. The displayed field automorphism splits the extension and proves $\mathrm{Cor}\,\Gamma = G \rtimes C_3$. This is the existential source Theorem 3.16, with its generation proof repaired. □

More generally, for every odd element order $m \equiv 2 \pmod 3$ the uniform common theorem gives a regular hypermap of type (m, m, m) with group G: let the three free-product involutions

act on the flag set G by right multiplication by a, b, c. Generation makes this action transitive, and its full commuting automorphism group is left multiplication by G. The field automorphism gives the triality. A type-permuting symmetry can be normalized to fix the identity flag and then acts on all words in the three generators; it induces the same marked group automorphism as above. The three-point argument therefore again excludes a duality, and the full type-permuting symmetry group is $G \rtimes C_3$. No unproved flag-transitivity of a general-m coset geometry is used for this hypermap conclusion. Exactly one of $q + \sqrt{2q} + 1$ and $q - \sqrt{2q} + 1$ is two modulo three: $q \equiv -1 \pmod 3$ and $\sqrt{2q}$ is 1 or -1 modulo three. This recovers the source's full-torus hypermap choices and its stronger existential assertion for every realized $m \equiv 2 \pmod 3$.

5 Corrections and retained background boundaries

Apply the written uniform theorem in $K = \mathrm{Sz}(8)$ with inverse field automorphism and $m = 5$, and embed it as the standard subfield of $G = \mathrm{Sz}(2^{15})$. The ambient automorphism F^5 restricts to F^2 on K. The triple therefore generates exactly K, not G, and has distinct rotation groups and a moved marked fixed point. Lemma 2 proves that its triangle is proper and nondegenerate. The source triangle orbit still has $|G|$ chambers, but its chamber graph has $[G : K] > 1$ components. This is an exact written counterexample to the unrestricted second assertion of Theorem 3.2, Corollary 3.5, the "any triangle" ambient-group part of Corollary 3.7, and Theorem 3.14 as printed. It also qualifies the final claim that ambient residual connectivity is straightforward. The positive replacements above require a selected fully generating seed; they do not identify the source's literal theorem set with a true theorem set.

For the source's geometric warning, $\mathrm{Stab}_G O(D_{10})$ has order twenty, not ten. Indeed five divides a nonsplit torus order. In its normalizer $T : C_4$ every involution is the square of an order-four element, because it lies in a Sylow 2-subgroup, which is cyclic of order four. Given the reflection r of D_{10}, choose such a square root x. The element x normalizes its unique C_5 and centralizes $r = x^2$, so normalizes D_{10} and preserves its support. Lemma 1 now gives the full stabilizer $C_5 : C_4$, of order twenty, isomorphic to $\mathrm{Sz}(2)$. Thus the source's reason for retaining chamber information rather than using the bare five-point subsets is exact.

The general chamber representation and connectivity facts in source Proposition 2.3 and Lemma 2.4 have elementary orbit proofs: send gB to gC and use its stabilizer B; panels are the corresponding $G(i)$-cosets; a word in the panel subgroups traces a gallery and every gallery gives such a word. Source Theorems 2.5 and 2.6 are explicitly retained general chamber/geometry correspondence theorems from Buekenhout–Cohen [2], not silently proved by our rank-three criterion. In the former, for a nontrivial Borel use $G(J) = \langle B, G(j) : j \in J \rangle$, in particular $G(\varnothing) = B$; the printed empty-generated-subgroup convention otherwise loses B. For example, $G = B = C_2$ with both panel subgroups equal to G gives a connected one-chamber, residually connected system. The printed identity with $L = \{0\}$, $J = \{1\}$, $K = \varnothing$ would say $G = 1$. Our application has $B = 1$ and directly proves the required rank-three correspondence, so no unrestricted presentation of that background theorem is needed as an additional application premise. A coset *pregeometry* is not called a geometry before the flag-extension condition is proved. The historical census numbers and earlier-priority remarks in the introduction remain attributed literature reports, not new enumerations or prerequisites.

References

[1] D. Leemans, K. Stokes and P. Tranchida, *Flag transitive geometries with trialities and no dualities coming from Suzuki groups*, arXiv:2311.13522v1, 22 November 2023, fourteen-page fixed edition. The source construction and all source-result attributions.

[2] F. Buekenhout and A. M. Cohen, *Diagram Geometry*, Springer, 2013, Propositions/Theorems 3.6.4, 3.6.7, 3.6.9 and 3.4.6, as cited in the fixed source. General background retained with the explicit Borel convention; no local complete external proof charge is known.